# Polynomial Kernel (of order 2)

***Example-2***

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$$

***Normal Model***

Calculate $a \cdot b$

$$\begin{bmatrix} a_1 & a_2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

***Map Input data from 2-D to 6-D***

$$\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$$

**Vectors Mapping**

$a, b \rightarrow a, b$

$$\begin{bmatrix} a_1^2 & b_1^2 \\ a_2^2 & b_2^2 \\ \sqrt{2}a_1a_2 & \sqrt{2}b_1b_2 \end{bmatrix}$$

***Kernelized Model***

Calculate $a \cdot b$

$$\begin{bmatrix} a_1^2 & a_2^2 & \sqrt{2}a_1a_2 & \sqrt{2}a_1 & \sqrt{2}a_2 & 1 \end{bmatrix} \begin{bmatrix} b_1^2 \\ b_2^2 \\ \sqrt{2}b_1b_2 \end{bmatrix}$$

*Higher Computational Complexity!!!*